

第五章 属性文法 1-1 课后作业.

1. 写出下面文法的 属性文法

$$N \rightarrow N_1 D \quad N.val = 10 \times N_1.val + D.val$$

$$N \rightarrow D \quad N.val = D.val$$

$$D \rightarrow 0 \dots 9 \quad D.val = \text{lexval}(0 \dots 9)$$

1. 写出下面文法的 属性文法

$$N \rightarrow D N_1 \quad \begin{aligned} N.mul &= N.mul \times 10 \\ N.val &= N_1.val + N.mul \times D.val \end{aligned}$$

$$N \rightarrow D \quad \begin{aligned} N.val &= D.val \\ N.mul &= 1 \end{aligned}$$

$$D \rightarrow 0 \dots 9 \quad D.val = \text{lexval}(0, \dots, 9)$$

3. 十进制浮点数的文法修改如下

$$D \rightarrow N.S$$

$$N \rightarrow N_1 \text{ digit} \mid \text{digit}$$

$$S \rightarrow \text{digit} S_1 \mid \text{digit}$$

$$\text{digit} \rightarrow 0 \mid 1 \mid 2 \mid 3 \mid 4 \mid 5 \mid 6 \mid 7 \mid 8 \mid 9$$

$$D \rightarrow N.S \quad D.val = N.val + S.val$$

$$N \rightarrow N_1 \text{ digit} \quad N.val = 10 \times N_1.val + \text{digit}.val$$

$$N \rightarrow \text{digit} \quad N.val = \text{digit}.val$$

$$S \rightarrow \text{digit} S_1 \quad S.val = 0.1 \times (\text{digit}.val + S_1.val)$$

$$S \rightarrow \text{digit} \quad S.val = 0.1 \times \text{digit}.val$$

$$\text{digit} \rightarrow 0 \dots 9 \quad \text{digit}.val = \text{lexval}(0 \dots 9)$$